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Predicting car park availability for a better delivery bay management

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Abstract

Freight parking is a serious problem in smart mobility and we address it in an innovative manner. We propose an intelligent urban parking management system capable to modify in real time the status of any parking spaces, from a conventional place to a delivery bay and inversely. Its driving force is the parking availability prediction. In this article, we model the parking occupancy by many regression types. We analyze their results, identify their strengths as well as their weaknesses and deduce the most effective one. We calculate the probability of a place being left free by the actuarial method. To do so, we exploit a massive amount of real-time parking availability data collected and disseminated by the City of Melbourne, Australia.

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1. Introduction

Double-parking is a bad practice incrementally adopted by carriers in order to avoid any deterioration in their performance. This illegal behavior generates congestion and thus road insecurity, additional noise, air pollutants but also supplementary costs. Furthermore, the growth of urban population, the emergence of new delivery modes

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(home delivery, pick-up points and lockers), the increase in the number of e-commerce platforms and the restriction of traffic areas complicate additionally logistics operators' delivery tasks. As a consequence, municipalities implement free delivery bays in zones where demand surpasses supply. In contrast, if their spatial configuration (number, location and size) is inadequate, their organization (full-time / part-time) is sub-optimal or their management rules compliance are not clear, the drivers continue to park on prohibited places to improve their productivity.

In like circumstances, local authorities worldwide, especially in large cities and metropolises, come under pressure to increase the number of delivery bays [1]. From another angle, this action implies a reduction in public space and private car parking because of the limited size of the city. It therefore conduces to an unbalanced sharing of these resources in favor of transport operators. Aware of this stake, we suggest an intelligent urban parking management system. It is able to recognize, understand, communicate with its environment, learn and make decisions. It predicts, in real time, the occupancy of a city's parking spaces in order to alter their status dynamically from a delivery area to an ordinary place and vice versa. Its aim is to assign the most suitable places, in terms of location and walking distance, to deliverers without penalizing other road users. It allows to interactively book one or multiple places, to finalize a pre-established delivery tour and to construct, from scratch and upstream, the itinerary to follow. Also, it automatically reserves spots for carriers opening their information systems, when they approach their destination.

Our solution is heterogeneous and consists of two modules. The first predicts, in real time, its managed spaces' availability and this article is dedicated to this module's development. Its outputs serve as inputs for the second module that allocates the free spaces to the carriers requesting them. In this paper, we investigate a variety of forecast methods (Linear regression, Gradient boosting, Random forest, Neural networks...) to analyze a parking lot's occupancy historic and predict its future availability. The idea is to find the method both minimizing the necessary computation costs and maintaining high precision. To boost our results' reliability, we estimate also the probability of a parking place remaining available in the upcoming time intervals by means of the survival analysis.

The rest of this document is organized as follows: section 2 reviews the state-of-the-art in parking availability prediction, section 3 discusses our system architecture and describes the prediction techniques employed, section 4 introduces the real database utilized and evaluates our results, section 5 presents the probabilities calculated by the actuarial method and section 6 outlines conclusions and future work.

2. Related works

In recent years, parking availability prediction, although indicated as a challenge, is excessively acknowledged and largely documented. It is closely linked to multiple factors interaction such as parking patterns, parking rules, parking facilities type, parking pricing policy, driver behavior, conductor preferences...

Numerous papers sought the most performing technique by comparing several and different methods. An approach for predicting parking spaces availability is defined in [2]. The authors modeled the occupancy rates employing three non-parametric algorithms (neural networks, regression tree and support vector machines) with three different time-series feature sets. The evaluation reveals that the regression tree with a feature set containing the history of observations, time of day and day of week provides the best performance. Stolfi et al. [3] studied various prediction strategies (Fourier series, k-means clustering, polynomials fitted to the k-means' centroids...) to analyze the historical occupancy rates of car parks and forecast the future availability. Their experiments disclose that the time series predictor is the most efficient.

However, some works attempted to reach a highly technical level by focusing on a single prediction model. In this sense, Camero et al. [4] developed a new deep learning technique based on recurrent neural networks (RNN) to estimate the number of free parking lots near to the medium time horizon. They defined an optimal architecture of the RNN by exploiting two metaheuristics (genetic algorithm and (1+1) evolutionary strategy). They compared their predictions against the ones presented in [3]. Their results show that the performance of their proposal does not exceed all its competitors and its forecast values are similar to polynomials or time series. In [5], the parking lot is modeled by a continuous-time Markov chain, following Caliskan et al. [6], for the purpose of predicting future occupancies in an ad hoc network. Nevertheless, this method can only be effective for car parks that have an admission control such as a gate and monitor the occupancy of the parking space. To overcome this flaw, a hybrid

approach combining agent-based model (ABM) and Markov chain Monte-Carlo (MCMC) is proposed in [7]. It forecasts transportation patterns and parking lot utilization. It initializes the MCMC proposal distribution with ABM data. Therefore, it significantly reduces the prediction error over standard MCMC and also improves upon the ABM alone. A systematic framework for multiple steps ahead parking availability prediction is depicted in [8]. It encompasses two modules. The first introduces neural networks for the short-term parking occupancy prognosis in selected regions of an urban road network, while the second applies survival analysis to forecast the probability a free space to continue being free in subsequent time intervals. Their findings point out that the duration of free parking space follows a Weibull distribution.

Other authors dealt with this issue from further perspectives. For instance, Richter et al. [9] addressed it using historic data, with a special focus on how to reduce the number of required models. Thus, they utilize five different clustering strategies (a one-day model, a three-day model and a seven-day model per road segment, temporal clustering and spatial clustering). Their results show that thanks to clustering, a dramatic reduction of storage requirements (up to 99%) can take place, while still achieving good accuracy of the predictions.

In our turn, we test the performance of simple methods, easy to implement and convenient to parameterize (Dummy regression and Linear regression). They are very little applied regression techniques in parking modeling. We evaluate their reliability against that of more complex ones, namely Decision tree, Gradient boosting, Random forest, K-nearest neighbors and Artificial neural networks. Then, we combine these techniques using stacking model to improve the prediction accuracy. We look for a compromise between the estimator's intricacy (needed storage, training parameters' number and time cost) and its precision, hence the selection of rising complex methods.

3. Methodology

The proposed parking occupancy prediction system is modelled on the stacking algorithm with two levels. It considers $x = \{id, t, d\}$ as input and generates $y = A(id, t, d)$ as output, where id is the area name, t is the time, d is the day of week and $A(id, t, d)$ is the number of available places in area id at time t of day d .

$$A(id, t, d) = T(id) - O(id, t-1, d) - E(id, t, d) + D(id, t, d) \quad (1)$$

$O(id, t-1, d)$ is the number of occupied places at $t-1$, $E(id, t, d)$ is the number of incoming vehicles at t , $D(id, t, d)$ is the number of departing vehicles at t and $T(id)$ is the total number of places in the area id .

Stacking or super Learner is a loss-based supervised learning method which combines many candidate learners to build an optimal estimator [10]. The idea is to create a meta-dataset containing a tuple for each tuple in the original dataset. However, instead of using the original input attributes, it uses the predictions of the base learners as the input attributes. The target attribute remains as in the original training set.

In the present study, we use a set of six base learning algorithms $\{M_1, M_2, \dots, M_6\}$ and a meta learner, each of which belongs to a different algorithm category and requires a specific set of parameters. Their internal principles, that are very dissimilar, are described below.

Dummy regression is a baseline regression model. For any input data, it predicts either the mean of the training set, its median, its quantile or a constant value provided by the user. The mean strategy is the most appropriate to meet our needs, so we apply it.

In contrast to Linear regression, which establishes a single linear forecast formula valid over the entire data space [11], Regression Tree subdivides it, via binary recursive partitioning, into a series of rectangular sets gathering the most homogeneous collection of outcomes possible. Then, it fits a simple prediction model within each smaller region [12]. It starts with one node containing the whole sample. The parent node can only be branched into exactly two descendent nodes and each child node will, in turn, become a parent node, unless it is a terminal node [13]. The variable independent x_i and the split point s are chosen to minimize the sum of squared error between the observation and the mean in each node (i.e., maximize node homogeneity or underplay node impurity). The prediction y_c , for leaf C of tree T , is equal to the sample mean of the dependent variable in this cell.

The tree building process goes on until it is impossible to continue. It is stopped when the minimum value of the splitting rule, maximum number of leaves or minimum number of data points in the terminal nodes is reached [14]. In our case, we use, as a stopping criterion, the parameter `max_depth`. We set its value at 10 leaves maximum.

A concern with growing trees completely founded on the use of stopping rules is that important associations may be missed because of stopping too soon [15]. To avoid tree under-fitting and over-fitting while keeping a reasonable computation time, a cost-complexity pruning algorithm is also implemented. It computes a set of consecutive nested sub-trees T_i , $i \in \{1, 2, \dots, R\}$, of decreasing size from the original large tree T_0 by progressively pruning upwards to its root node T_R [16].

Gradient boosting is an ensemble predictor. It sequentially produces a linear combination of simple predictors, in our case decision trees, by solving an infinite-dimensional convex optimization problem [17]. The principle idea behind this algorithm is to construct the new base-learners to be maximally correlated with the negative gradient of the loss function, associated with the whole ensemble [18]. The loss function applied is the classic squared-error loss. To perform this technique, we use an open source library called XGBoost (eXtreme Gradient Boosting). It is an accurate and scalable implementation of gradient boosting framework.

One widely used non-parametric method is the k-nearest neighbor (KNN) method. In KNN, the dependent variable is predicted as a weighted mean of k nearest observations in a database, where the nearness is defined in terms of similarity with respect to the independent variables of the model. We choose a size neighborhood equal to 10, opt for the standard Euclidean metric and adopt the distance-weighted function [19].

An Artificial neural network is a mathematical model that tries to simulate the structure and functionalities of biological neural networks [20]. It is composed by several hidden layers of neurons where the output of a neuron of a layer becomes the input of a neuron of the next layer. An artificial neuron $y_i = f_i(x) = \varphi(\langle w_i, x \rangle + b_i)$ is a function f_i of the input $x = (x_1, \dots, x_m)$ weighted by a vector of connection weights $w_i = (w_{1,i}, \dots, w_{m,i})$, completed by a neuron bias b_i , and associated to an activation function φ [21].

To train our multilayer perceptron, we use the Rectified Linear Unit (ReLU) activation function that returns $\varphi(x) = \max(0, x)$ and the stochastic gradient-based optimizer proposed in [22] as a solver for the log-loss function.

Random forest is an ensemble learning method that corrects for each individual tree's propensity to overfit the training set. It employs bagging method and a CART-like tree learning algorithm in order to build K de-correlated decision trees. These trees that are being trained on each individual bootstrapped dataset are generally not pruned. Instead of considering all m predictors at a split, it uses only a randomly selected subset of the m predictors [23].

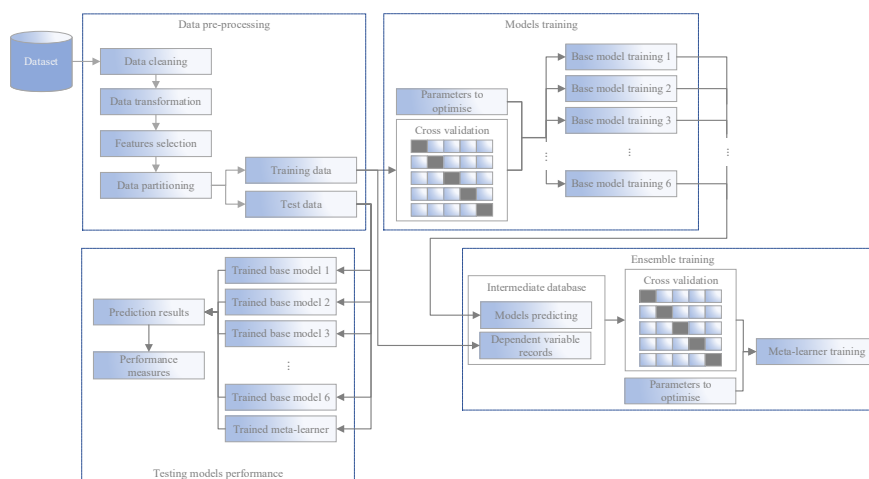


Fig. 1. The proposed parking occupancy prediction system architecture.

Our solution's methodological framework (Fig. 1) consists of four major phases (Data pre-processing, Models training, Ensemble training and Testing models performance), each of which includes several steps. The main ones are as follows:

- Pre-process the database and split it into two distinct subsets: 80% for training and 20% for testing;
- Divide the training set at random into 10 equal-size folds by means of cross validation;
- Learn each base model M_u (dummy regression, linear regression, regression tree...) on the training set, generate its n cross-validated predictions a_{iu} and combine these values to form a matrix $A = (a_{iu}) \in \mathbb{R}^{n \times 6}$;
- Fit the meta-learner model (random forest) on the new dataset containing the matrix A and the label attribute vector;
- Evaluate all learning techniques by making predictions on the test data and calculating three error metrics (coefficient of determination R2, mean absolute error MAE and root mean squared error RMSE).

4. Implementation and Results

The proposed approach is implemented in python 3 under the JupyterLab web development environment. It is tested by exploiting the parking history in downtown Melbourne, Australia.

4.1. Data set: Melbourne On-street Car Parking Occupancy

The City of Melbourne has installed approximately 4 300 in-ground sensors covering 303 street segments in 35 areas across the central business district. These sensors record the entry and exit time of a vehicle. Data features include the arrival time, departure time, duration of parking events (in seconds), area name, street name...

We analyze the historic data for the 2017 calendar year available from [24]. In total, the database contains 35 908 877 parking event records. We process the data to structure them, complete missing ones and remove noisy and inconsistent values falsifying the results. To this end, we carried out several remedial actions:

- Delete lines with very short (less than 3 minutes) or negative parking durations, remove unnecessary columns and group data by area;
- Assign a different integer between 0 and 34 to each area name;
- Extract the weekday from the arrival and departure time variables and attribute to each day an integer from 0 (for Monday) to 6 (for Sunday);
- Sample the timeline at 15-minute intervals and assign to each time lapse an integer from 0 (for the time range between 12 am and 12:15 am) to 95 (for the time range between 11:45 pm and 12 am);
- Complete registrations where the arrival or departure time is missing by midnight of the arrival or departure day respectively;
- Calculate the parking capacity of each area by counting as many sensors as installed in it;
- For each area, determine the number of incoming (respectively outgoing) vehicles during each time window by counting the number of arrival (respectively departure) times recorded within this interval;
- Deduct the number of occupied places in a time period from these last two measurements and the number of busy spaces at the previous range;
- For each area and time slot, obtain the number of spaces available by subtracting the number of those occupied from its parking capacity;
- If, for a given time slot, no data exist, copy the value of either the preceding time slot or the same previous day's slot (the same day of the previous week).

After processing the data, our data set consisted of 1 216 813 occupancy values for 4 300 parking spaces over 365 days. Figure 2 shows the annual parking demand for each area and its capacity. We can remark that the area capacities vary from 42 to 2700 parking spaces, i.e. there are some areas where the parking supply is important (e.g. area 3, 4 and 9) and others where it is very mediocre (e.g. area 31, 32 and 34). In general, we also note a great balance between the yearly number of parked cars in each area and its capacity, apart from areas 3 and 9, in which these two parameters are unbalanced.

Figure 3 displays the data breakdown of the car park availability on weekdays depicted as a bar chart. We can observe that the number of available places in each area is quite similar for all weekdays. Conversely, we notice that

it is more important in areas 3, 4 and 9 than in the other ones, which is expected since they offer the most parking spots.

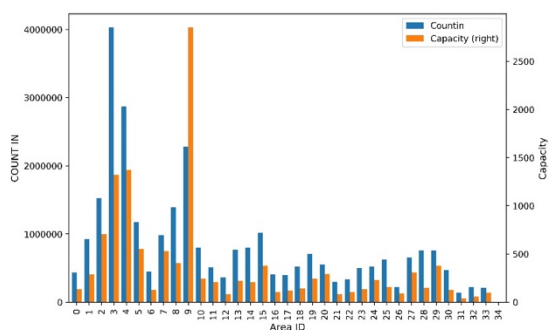


Fig. 2. The parking capacity and annual demand for each area.

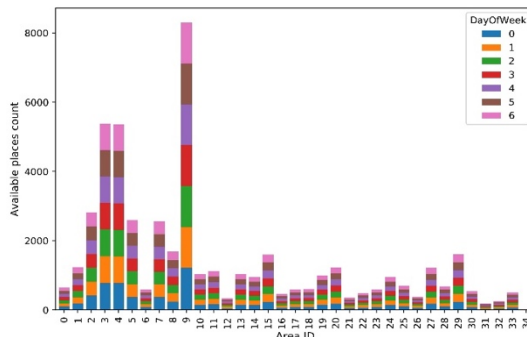


Fig. 3. Daily distribution of car park availability.

4.2. Experimental results

By analyzing the three error metrics calculated, it is clear that the results of all methods without exception far exceed those of the two basic predictors (Linear regression and Dummy regression), which is expected because they do not take into account the relationship between the variables and make strong hypothesis about their distribution. The artificial neural network is ranked third despite its long computation time, increased algorithmic complexity and large storage memory required. It took more than 6 hours to run while the other methods only needed a maximum of 12 minutes. Scoring the lowest MSE value (21.65677), Random forest approach performed best, followed by Stacking, XGboost and Regression tree with performance drops of 1.87%, 2.62% and 5.41% respectively. From MAE, the most effective method having the lowest error metric (2.16695) is Random forest that is 1.69% better than Regression tree, 2.10% better than XGboost and 2.17% better than Stacking. In terms of R2, the most accurate method is Random forest with a coefficient of 0.87976 surpassing Stacking by nearly 0.003%, XGboost by almost 0.4% and Regression tree by about 0.8%. Overall, the Random forest method outperforms each of the individual models and even the ensemble method regarding its excellent results and short execution time.

Figure 4 illustrates a graphical representation of the estimates based on the true values. The awaited curve is a straight line passing through the reference point origin. Most forecasts provide large numbers of available spaces, this may be due to the data set used, which tends to contain a high concentration of the large numbers of available spaces. For most regression models tested, predictions are more accurate when numbers of available spaces are small ($y_{test} \leq 50$).

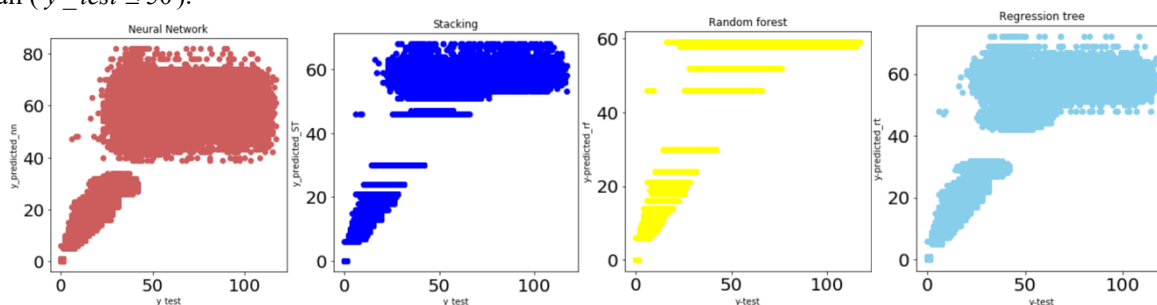


Fig. 4. Comparative graph between the predictions obtained by some techniques and the real values.

The suggested predictive methodology allows to achieve very good results with determination coefficients close to 1 (≈ 0.88) and absolute errors not exceeding 3 places. It is supplemented by survival probabilities calculation to significantly improve real-time operational efficiency of parking space management.

5. Survival Analysis of parking space availability

The number of available spaces is a necessary parameter but not sufficient to properly manage the urban parking system. An empty space can be suddenly filled just before the carrier's expected arrival. In order to estimate this risk, we calculate the probability of a place remaining available by the actuarial method of survival analysis.

This nonparametric approach estimates survival rates by calculating conditional probabilities over a priori fixed time intervals. These intervals are of equal length and it is 15 minutes for our case. Survival function $\hat{S}(j)$ is obtained by multiplication of the successive conditional probabilities [25].

$$\hat{S}(j) = \prod_{k=1}^j 1 - \frac{2 \cdot d_k}{2 \cdot N_k - m_k} \quad (2)$$

Where d_k is the number of places occupied during interval $I_k = [t_k, t_{k+1})$, m_k is the number of censored spell endings observed in interval I_k and N_k is the number of places vacant at start of interval.

Hazard-based parking space availability duration models are developed for each parking area. Several available space survival curves (Figure 5) are analyzed, with independent variable being the type of day (weekend/weekday). The observation of these curves induces interesting results concerning the continuity of parking space availability in the coming intervals.

In area 5, the probability that a place will remain available after 5 minutes on Monday is 0.80 and on Sunday is 0.91. The probability of a place in the area 15 being left available after 8 minutes on Monday is 0.78 and passes to 0.69 on Sunday. For the area 25, the probability for a spot to stay free after 10 minutes on Monday is 0.72 and 0.81 on Sunday. The survival probability tends towards 0 when the durations exceed 170 min in the area 5, 70 min on Sunday versus 210 min on Monday in the area 15 and 70 min on Monday against 260 min on Sunday in the area 25. It should be added that there is a significant difference between the survival probabilities during weekdays and weekends. Tertiary and administrative areas (5, 25, 33...) are characterized by higher survival probabilities on weekends. Whereas tourist and commercial zones (15, 29, 32...) are distinguished by higher probabilities on weekdays.

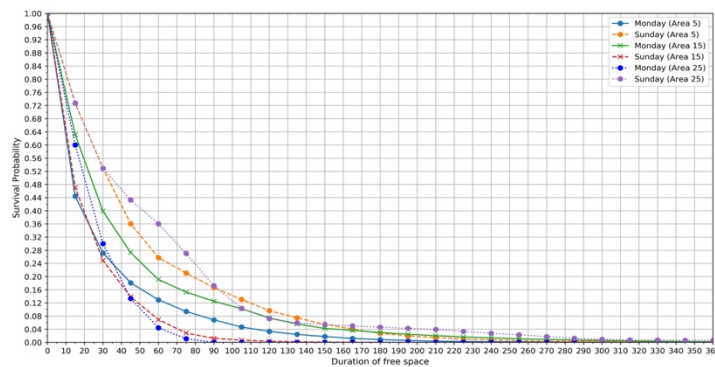


Fig. 5. Survival curves of space availability in three areas (area 5, 15 and 25).

6. Conclusions and Future works

In this article, we propose a new tool managing the urban parking system able to pre-book spaces, organize carrier trips and channel driver behavior. It not only streamlines parking infrastructures but also eliminates several traffic movements on major road sections. It is constructed in two modules: the first, implemented above, predicts the parking availability and the second allocates the free spaces to the carriers.

Our tool's efficiency and performance are conditioned by the drivers' ability to respect their reservations and arrive on time. This is why it is essential to strengthen the proposal with additional services (blacklisting users posing a threat, disturbance management system, fine imposition and warning generation facility...). In the future, we will develop the affectation module based on mathematical programming, implement complementary functionalities to ensure the smooth running of urban parking and evaluate our solution's contribution compared to the ordinary system.

References

- [1] Comi, Antonio, Schiraldi Massimiliano M., and Buttarazzi Berta. (2018) "Smart urban freight transport: tools for planning and optimizing delivery operations." *Simulation Modelling Practice and Theory* **88**: 48-61.
- [2] Zheng, Yanxu, Rajasegarar Sutharshan, and Leckie Christopher. (2015) "Parking availability prediction for sensor-enabled car parks in smart cities." *2015 IEEE Tenth International Conference on Intelligent Sensors, Sensor Networks and Information Processing (ISSNIP)*, Singapore, pp. 1-6.
- [3] Stolfi, Daniel H., Alba Enrique, and Yao Xin. (2017) "Predicting car park occupancy rates in smart cities". *The Second International Conference on Smart Cities*, Málaga, pp. 107-117.
- [4] Camero, Andres, Toutouh Jamal, Stolfi Daniel H., and Alba Enrique. (2019) "Evolutionary Deep Learning for Car Park Occupancy Prediction in Smart Cities", in: Battiti Roberto, Brunato Mauro, Kotsireas Ilias and Pardalos Panos M. (eds) *Learning and Intelligent Optimization*. LION 12 2018. Lecture Notes in Computer Science, vol. 11353. Springer, Cham.
- [5] Klappenecker, Andreas, Lee Hyunyoung, and Welch Jennifer L. (2014) "Finding available parking spaces made easy." *Ad Hoc Networks (Elsevier)* **12**: 43-49.
- [6] Caliskan, Murat, Barthels Andreas, Scheuermann Bjorn, and Mauve Martin. (2007) "Predicting parking lot occupancy in vehicular ad hoc networks." *2007 IEEE 65th Vehicular Technology Conference - VTC2007-Spring*, Dublin, pp. 277-281.
- [7] Beheshti, Rahmatollah, and Sukthankar Gita. (2014) "A hybrid modeling approach for parking and traffic prediction in urban simulations." *AI Society* **30(3)**: 333-344.
- [8] Vlahogianni, Eleni I., Kepaptsoglou Konstantinos, Tsetsos Vassileios, and Karlaftis Matthew G. (2016) "A real-time parking prediction system for smart cities." *Journal of Intelligent Transportation Systems* **20(2)**: 192-204.
- [9] Richter, Felix, Martino Sergio D., and Mattfeld Dirk C. (2014) "Temporal and spatial clustering for a parking prediction service." *2014 IEEE 26th International Conference on Tools with Artificial Intelligence*, Limassol, pp. 278-282.
- [10] Cheng, Ju. (2016) *A General Ensemble Learning Framework for Online and Offline Machine Learning* (master thesis). University of California, Berkeley.
- [11] Shalizi, Cosma. (2006) *Lecture 10: Regression Trees*. Statistics 36-350: Data Mining.
- [12] Loh, Wei-Yin. (2011) "Classification and Regression Trees." *WIREs Data Mining and Knowledge Discovery* **1(1)**:14-23.
- [13] Moisen, G.G. (2008) "Classification and regression trees", in: Jorgensen Sven E. and Fath Brian D (eds) *Encyclopedia of Ecology*, vol. 1. Oxford, U.K.: Elsevier, pp. 582-588.
- [14] Breiman, Leo, Friedman Jerome, Stone Charles J., and Olshen R.A. (1984) *Classification and regression trees*. Chapman and Hall/CRC.
- [15] Lemon, Stephenie C., Roy Jason, Clark Melissa A., Friedmann Peter D., and Rakowski William. (2003) "Classification and regression tree analysis in public health: Methodological review and comparison with logistic regression." *Annals of Behavioral Medicine* **26(3)**: 172-181.
- [16] Prodromidis, Andreas L., and Stolfo Salvatore J. (2001) "Cost Complexity-Based Pruning of Ensemble Classifiers." *Knowledge and Information Systems* **3(449)**: 449-469.
- [17] Gérard, Biau, and Benoît Cadre. (2017) *Optimization by gradient boosting*. arXiv:1707.05023
- [18] Natekin, Alexey, and Knoll Alois. (2013) "Gradient boosting machines, a tutorial." *Frontiers in Neurobotics* 7:21.
- [19] Haara A, & Kangas A (2012) Comparing K nearest neighbors methods and Linear regression-is there reason to select one Over the other?. *Mathematical and Computational Forestry and Natural-Resource Sciences* 4(1): 50-65.
- [20] Krenker, Andrej, Bester Janez, and Kos Andrej. (2011) "Introduction to the artificial neural networks", in: Suzuki Kenji (eds) *Artificial Neural Networks: Methodological Advances and Biomedical Applications*. InTech, pp. 1-18.
- [21] Neural Networks and Introduction to Deep Learning, available at <https://www.math.univ-toulouse.fr/~besse/Wikistat/pdf/st-m-hdstat-rnn-deep-learning.pdf> and consulted on 16/09/2019.
- [22] Kingma, Diederik, and Ba Jimmy. (2014) *Adam: A Method for Stochastic Optimization*. arXiv:1412.6980.
- [23] Timothy, C. Au. (2017) "Random Forests, Decision Trees, and Categorical Predictors: The "Absent Levels" Problem." *Journal of Machine Learning Research* **19(1)**: 1737-1766.
- [24] On-street Car Parking Sensor Data – 2017, available at <https://data.melbourne.vic.gov.au/Transport-Movement/On-street-Car-Parking-Sensor-Data-2017/u9sa-j86i/data> and consulted on 19/09/2019.
- [25] Jenkins, Stephen P. (2004) *Survival Analysis*. Unpublished manuscript, Institute for Social and Economic Research, University of Essex, Colchester, UK. Downloadable from <http://www.iser.essex.ac.uk/teaching/degree/stephenj/ec968/pdfs/ec968lnotesv6.pdf>.